Baker College | 115 YEARS OF IMPACT

Screw Threads and Thread Gauges

Baker College Engineering

Brad Peirson
College of Information Technology and Engineering

Agenda

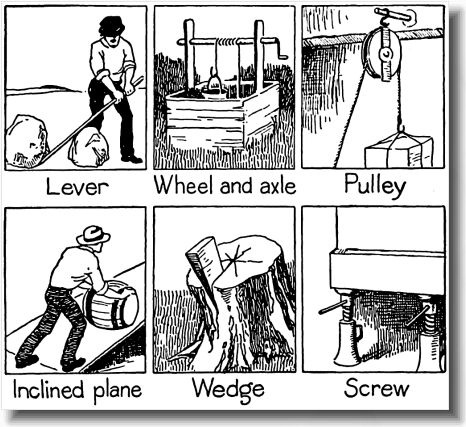
Thread Standards

Thread Types

Thread Forms

Thread Gauges

Simple Machines



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The screw is one of six *simple machines*–the basic machines on which all other machines are based.

The Archimedes Screw

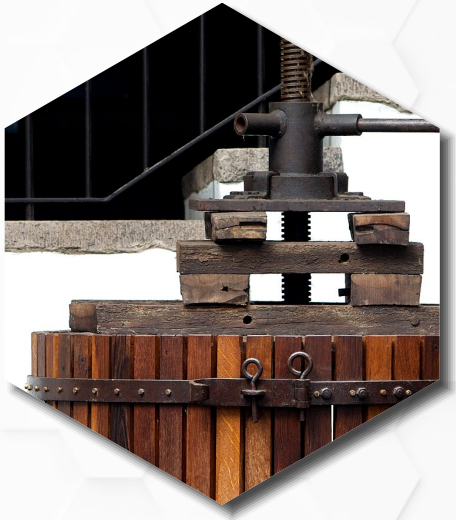
- One of the oldest and simplest machines
- Dates back to at least Archimedes (~220 B.C.) who used it to pump water
- Some historians believe the Egyptians were using it earlier



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Further Development

- Screw press used for squeezing grapes and olives
- DaVinci used screws in many inventions–invented the buttress thread



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Thread Standards

The Wild West of Screw Threads



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For a *very* long time craftsmen would use whatever thread shape suited them–as on these post medieval hooks.

The Whitworth Thread



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In 1841 Sir Joseph Whitworth created the first standard *thread form*.

Standardization of Screw Threads (1 of 2)



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Today, the metric world is on the ISO thread standard.

Standardization of Screw Threads (2 of 2)



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The U.S. standardized on American Standard threads for a long time. That standard was eventually supplemented by the Unified Standard thread. In 1949, American Standard was officially obsoleted and was wholly replaced by Unified Standard.

The Various Thread Standards

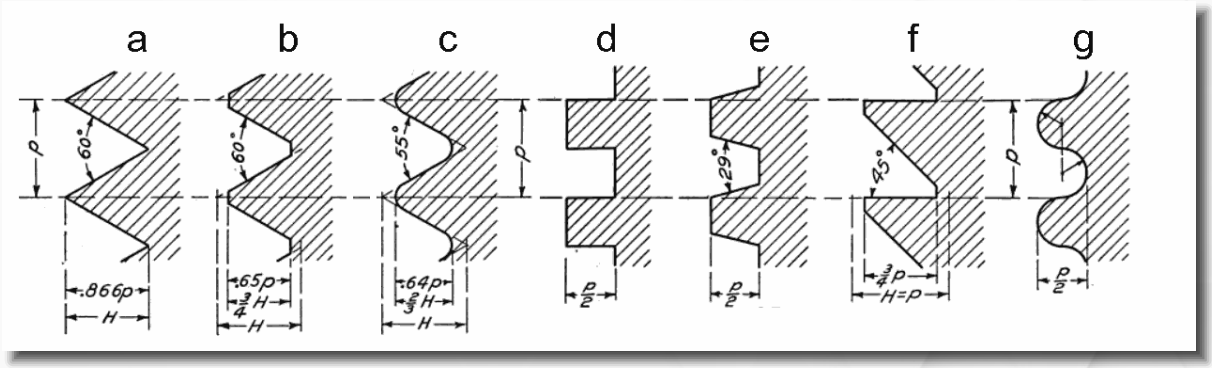


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Left to right: M12, 40mm LH – M12, 40 mm – M12x1.5, 40mm – M12x1.25, 40mm – UNF 1/2-20x1.5 – UNC 1/2-13x1.5 – BSW 1/2-12x1.5 – BSF 1/2-16x1.5

Thread Types

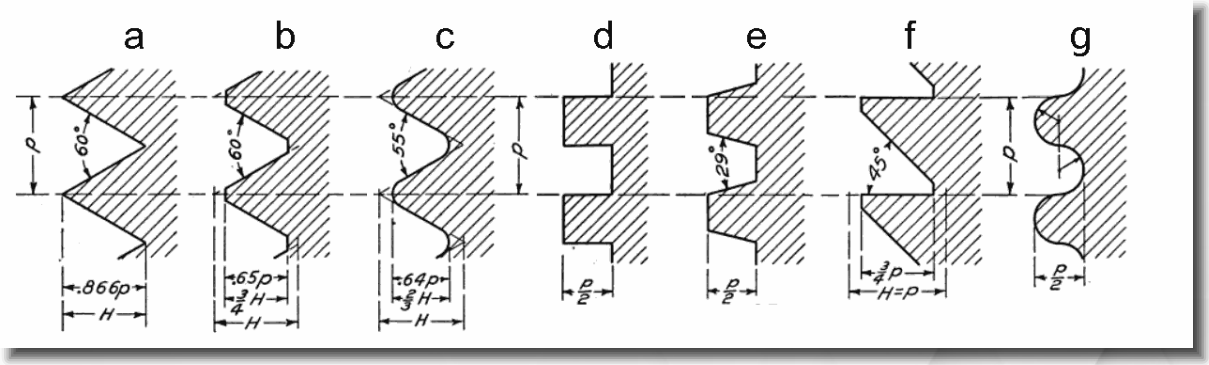
Types of Threads (1 of 2)



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Left to right: V-thread – American National Thread – Whitworth – Square – Acme – Buttress – Knuckle

Types of Threads (2 of 2)

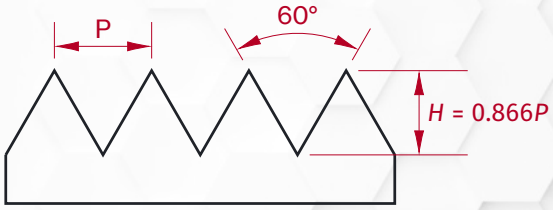


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Threads all serve the same basic purpose: as they rotate they (or what they're attached to) moves linearly. While they all serve that same purpose, not all types of threads are good at, or even capable of, the same jobs.

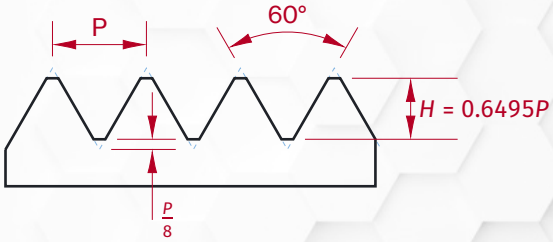
V Threads

- Early attempt at a standardized form
- Basic 60°“V” shape



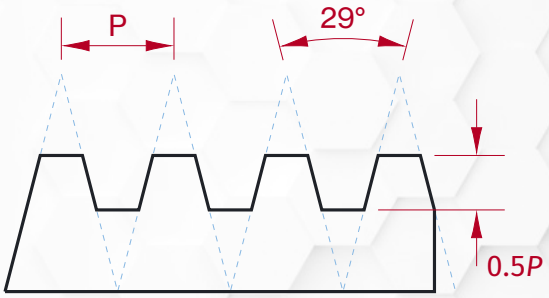
American National

- Standard US thread before Unified
- Same basic standard as V thread and Unified Standard
- General fastening



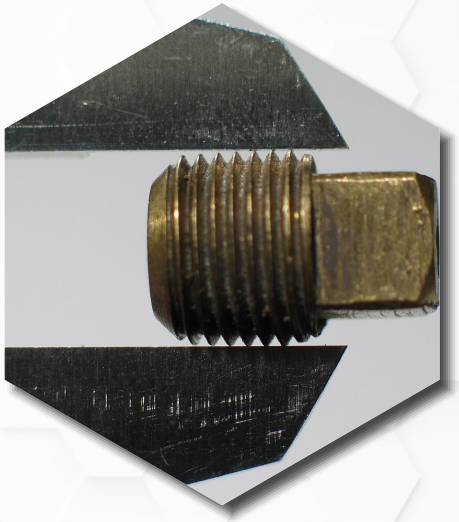
Acme Thread

- Thicker threads than most other designs
- Used strictly for linear motion



British Standard Pipe Thread

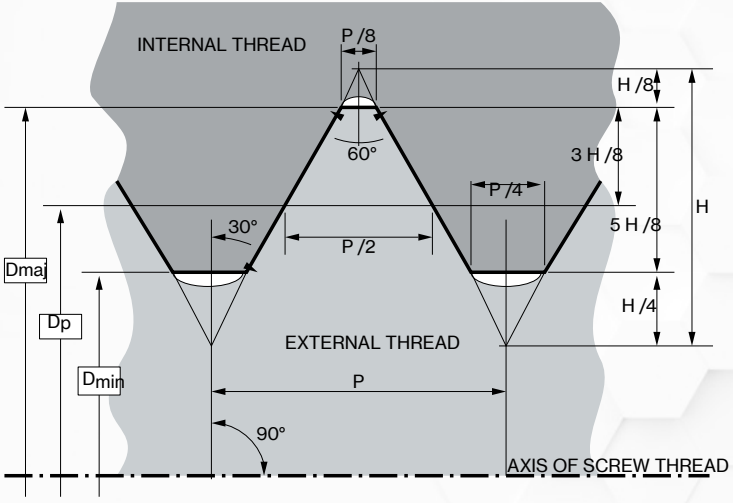
- Sides are not parallel – taper inward
- Gap between male and female gets smaller when tightened
- Designed to seal air/water



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Thread Forms

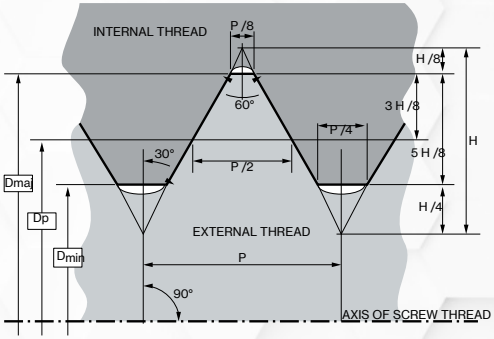
Thread Forms (1 of 3)



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Thread Forms (2 of 3)

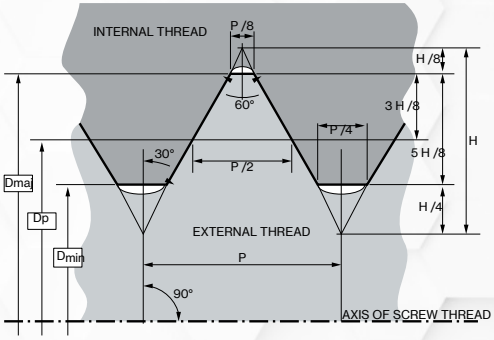
- A *thread form* is a technical drawing defining the shape of the threads
- Thread forms in modern standards are generic–dimensions are formulas



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Thread Forms (3 of 3)

- Basic dimensions based on **standard sizes**
- Other dimensions use the standard sizes in their forumlas

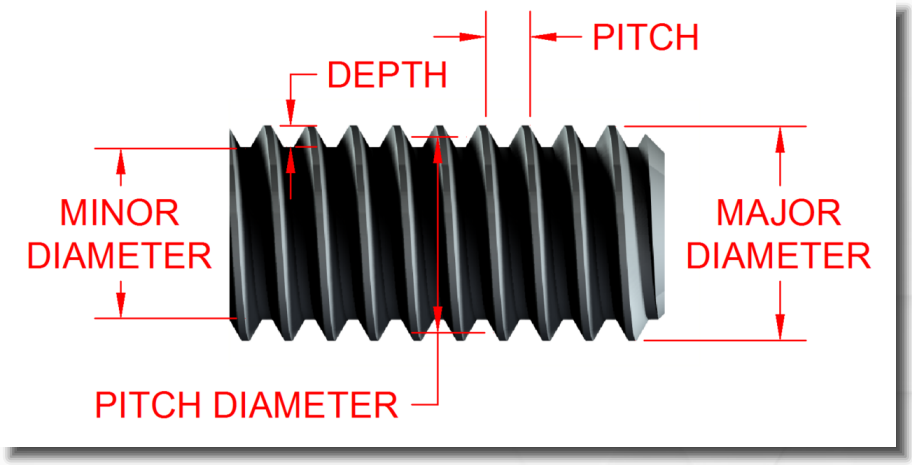


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Thread Forms

Thread Definitions

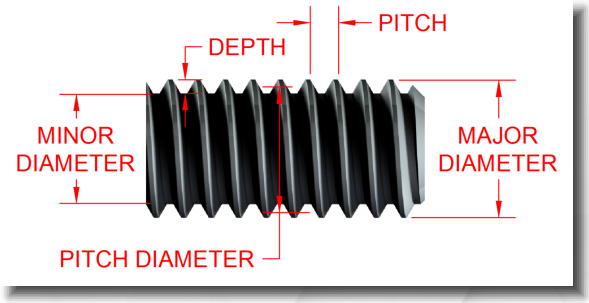
Thread Definitions



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Screw Thread

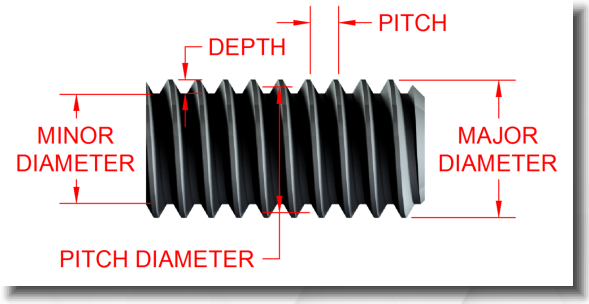
A ridge of uniform section in the form of a helix on the external or internal surface of a cylinder, or in the form of a conical spiral on the external or internal surface of a cone.



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Internal Thread

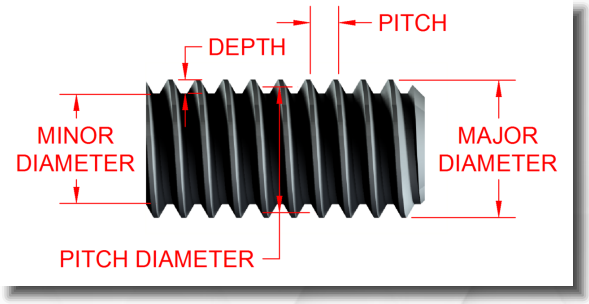
Thread on the inside of a member (nut)



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External Thread

Thread on the outside of a member (screw)



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Male and Female

- We really shouldn't use male and female to describe internal and external threads
- And not just out of fear of getting cancelled...
- Depending on the standard, the "male" may have internal threads

Luer Lock



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The *male* luer lock connector, common in syringes, uses *internal* threads. Because the threads don't do the sealing, the male taper inside the fitting does.

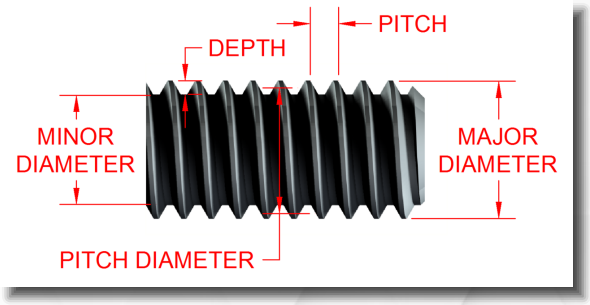
Thread Forms

Thread Dimensions

Thread Diameter

Major Diameter The largest diameter of the thread (screw *or* nut)

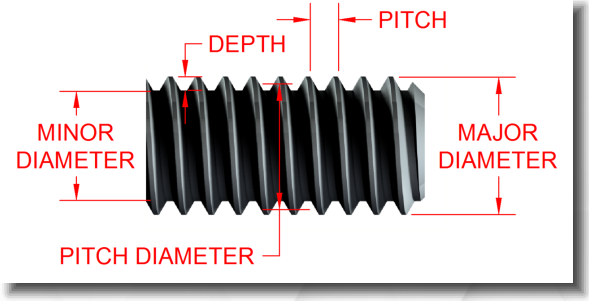
Minor Diameter The smallest diameter of the thread (screw *or* nut)



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Pitch Diameter

On a straight screw thread, the diameter of an imaginary cylinder, the width of which would pass through the threads at such points as to make equal the width of the threads and the width of the spaces cut by the surface of the cylinder

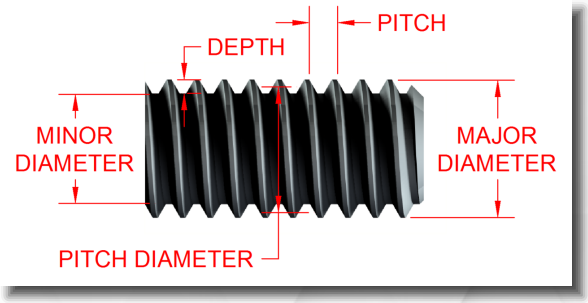


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Pitch

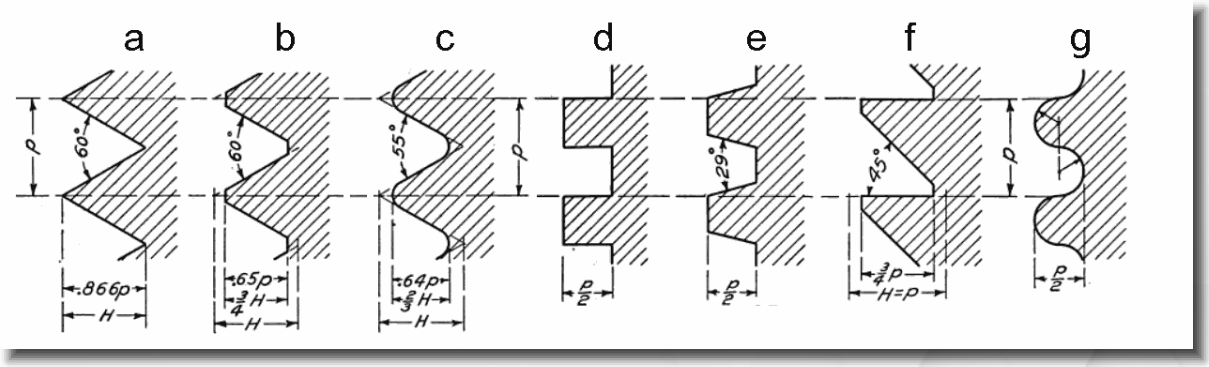
The distance from a point on the screw thread to a corresponding point on the next screw thread

$$\text{Pitch} = \frac{1}{\text{Threads/unit}}$$



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Pitch of Different Thread Types



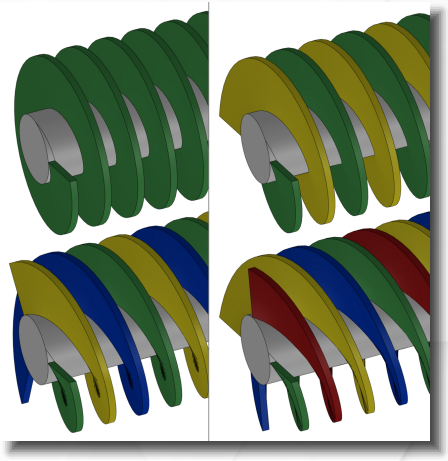
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While the *thread form* varies depending on the type of thread, the definition of *pitch* does not. This diagram shows the pitch of the 7 different thread forms shown.

Lead

The distance a screw thread advances axially in one turn

- Single start:
  Lead = Pitch
- Double start:
  Lead = 2 × Pitch
- Triple start:
  Lead = 3 × Pitch



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Screw Lead Video

Multiple Start Threads - SolidWorks 2017

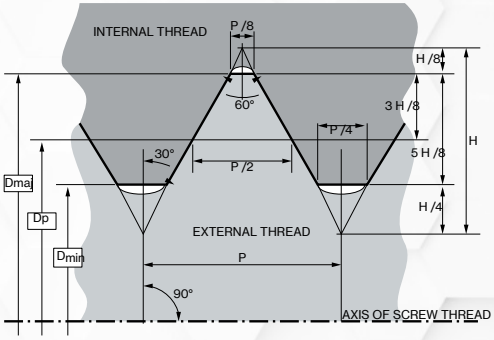
https://www.youtube.com/watch?v=73hLRT7n6H0

Video ©M5D / Cimquest via YouTube.com

Thread Angle

Angle of Thread The angle included between the sides of the thread measured in an axial plane

Half Angle of Thread The angle included between a side of the thread and the normal to the axis, measured in an axial plane



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Unified Thread Form Tolerances

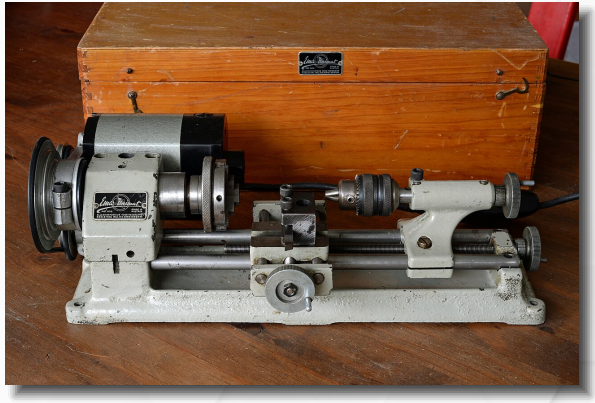
Nominal Size	External								Internal					
	Class	Allowance	Major Diameter			Pitch Diameter		UNR Minor	Class	Minor Diameter		Pitch Diameter		Major Diameter
			Max	Min	Min	Max	Min			Min	Max	Min	Max	Min
1/2-13 UNC	1A	0.0015	0.4985	0.4822	–	0.4485	0.4411	0.4069	1B	0.417	0.434	0.4500	0.4597	0.5000
	2A	0.0015	0.4985	0.4876	0.4822	0.4485	0.4435	0.4069	2B	0.417	0.434	0.4500	0.4565	0.5000
	3A	0.0000	0.5000	0.4891	–	0.4500	0.4463	0.4084	3B	0.4170	0.4284	0.4500	0.4548	0.5000
1/2-20 UNF	1A	0.0013	0.4987	0.4865	–	0.4662	0.4598	0.4392	1B	0.446	0.457	0.4675	0.4759	0.5000
	2A	0.0013	0.4987	0.4906	–	0.4662	0.4619	0.4392	2B	0.446	0.457	0.4675	0.4731	0.5000
	3A	0.0000	0.5000	0.4919	–	0.4675	0.4643	0.4405	3B	0.4460	0.4537	0.4675	0.4717	0.5625

Thread Forms

Creating Threads

Machining Threads

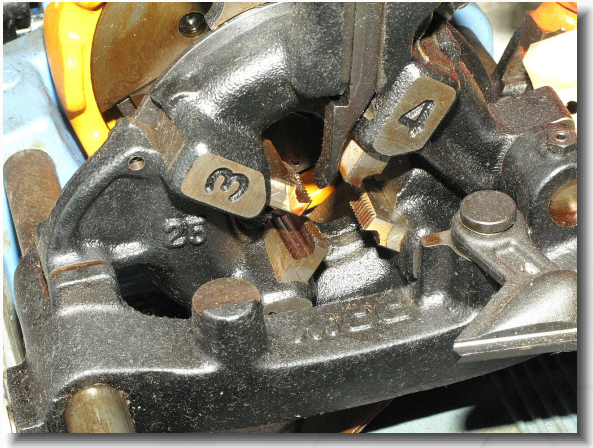
- Stock is chucked in a lathe
- The stock rotates, the lathe tool moves back and forth
- The lathe carriage is mounted on a screw
- The speed of the lathe screw determines the pitch of the machined screw



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Screw Machines

- Use a threading die (external) or collapsible tap (external)
- Stock is rotated in the tap/die cutting threads



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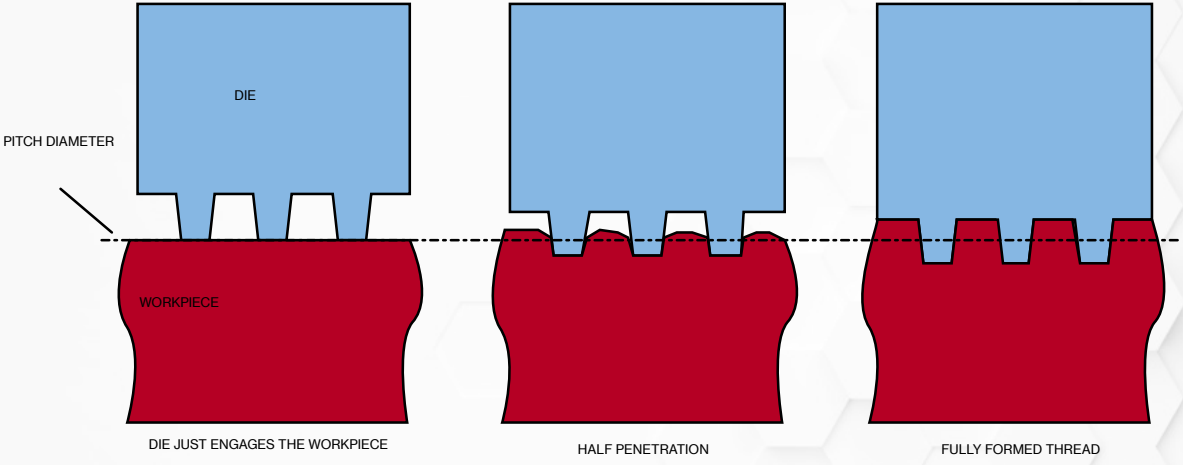
Thread Forming

- Most screw manufacturers *roll* or *form* threads
- This method is *much* faster and stronger than cutting
- The blank stock is rolled between dies–the material is deformed into threads



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Machining Threads

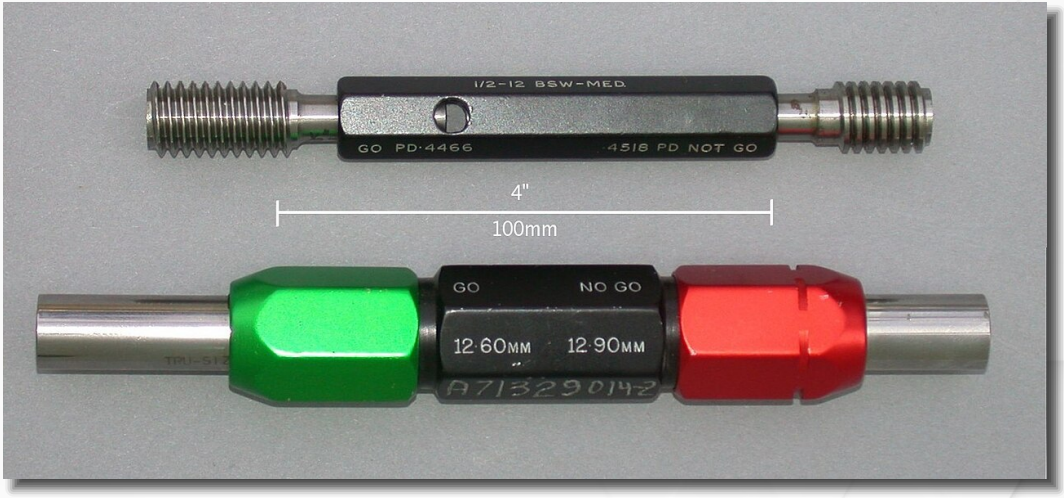


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Thread Gauges



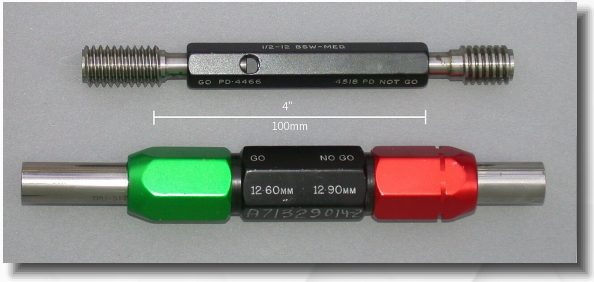
Go/No-Go Gauges



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Go Gauge (1 of 2)

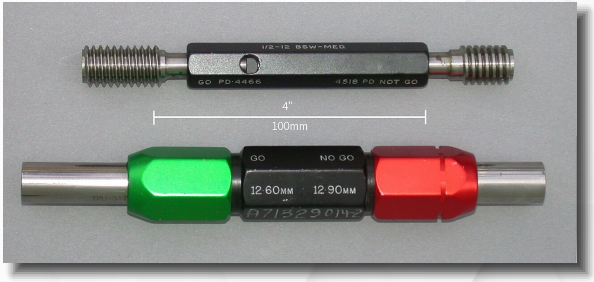
- Checks minor, pitch, major diameters
- OD ground to minimum minor diameter
- If gauge bottoms out thread is too shallow



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Go Gauge (2 of 2)

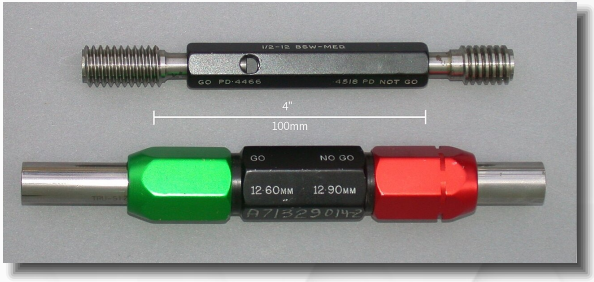
- Other dims being off will cause binding
- *Should not* bind at any point



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No-Go Gauge (1 of 2)

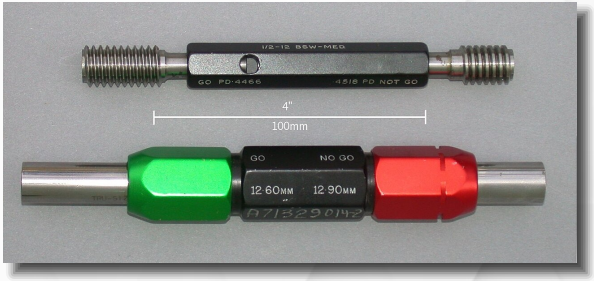
- OD ground to 2/3 major diameter
- Checks only pitch diameter
- Relieved in bottom of gauge to clear major diameter



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No-Go Gauge (2 of 2)

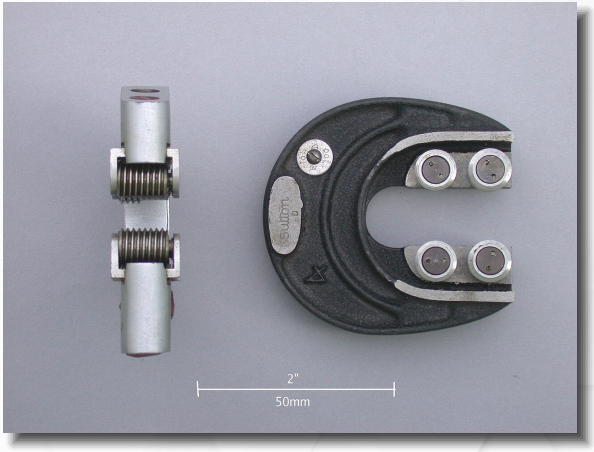
- Clears everywhere *but* pitch diameter
- It *should bind* which means pitch diameter is at least at minimum



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Thread Snap Gauges (1 of 2)

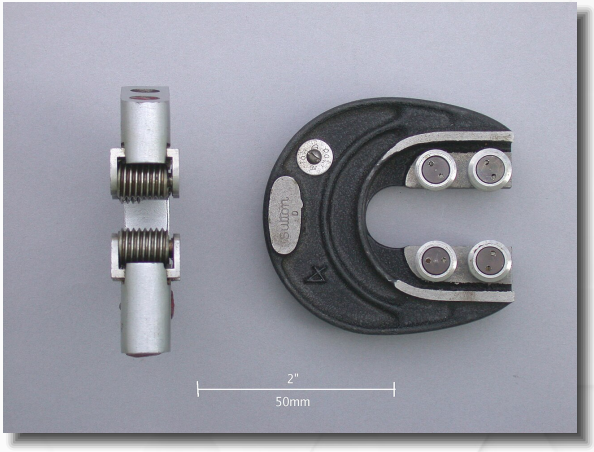
- Includes go and no-go jaws
- Can be fixed or adjustable for multiple screw sizes
- Jaws roll to minimize wear
- Measures more thread dimensions than standard plug go/no-go gauge



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Thread Snap Gauges (2 of 2)

- Screw slid from front (go) jaws to rear (no-go) jaws
- Should freely pass go jaw, bind on no-go jaws



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Thread Pitch Gauges (1 of 2)

- Measure thread profile *only*
- Really, can only reliably measure pitch
- May catch gross angle problems



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Thread Pitch Gauges (2 of 2)



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Questions?

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